

Introductory Calculus

ANSWER KEY



Direct substitution

- 1 Find $\lim_{x \rightarrow 2} (3x^2 - 2x + 1)$.
 $= 3(4) - 2(2) + 1 = 12 - 4 + 1 = 9$.
- 2 Find $\lim_{x \rightarrow -1} (x^3 - 5x)$.
 $= (-1)^3 - 5(-1) = -1 + 5 = 4$.

Indeterminate forms: factor and cancel

- 3 Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$.
Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 25}{x - 5} = \frac{(x - 5)(x + 5)}{x - 5} = x + 5$ (for $x \neq 5$), so the limit is $5 + 5 = 10$.
- 4 Find $\lim_{x \rightarrow -3} \frac{x^2 - 9}{x + 3}$.
Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 9}{x + 3} = \frac{(x + 3)(x - 3)}{x + 3} = x - 3$ (for $x \neq -3$), so the limit is $-3 - 3 = -6$.

The derivative as a limit

- 5 Write the difference-quotient definition of $f'(x)$.
$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$
- 6 Using Theorem 5.1 ($f(x) = x^2 \Rightarrow f'(x) = 2x$), find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.
 $f'(3) = 2(3) = 6$.
- 7 Using Theorem 5.1, find the slope of the tangent line to $f(x) = x^2$ at $x = -5$.
 $f'(-5) = 2(-5) = -10$.

Differential equations

- 8 Show that $y = x^2 - 7$ solves $\frac{dy}{dx} = 2x$.
By Theorem 5.1, the derivative of x^2 is $2x$, and the derivative of the constant -7 is 0 . So $\frac{dy}{dx} = 2x + 0 = 2x$, matching the equation exactly.

- 9 Explain why solving a differential equation gives a family of functions rather than a single number.

A differential equation only constrains the derivative, not the exact function; any constant added to a valid solution y has derivative 0 and doesn't change whether it solves the equation, so every $y =$ (that solution) $+C$ also works — a whole family, not one answer.

Multiple choice — easy

- 10 Find $\lim_{x \rightarrow 0} (-2x^2 - 6x - 6)$.

a. -6 ✓

b. 6

c. -10

d. -2

$$= -2(0)^2 - 6 = 0 - 6 = -6.$$

- 11 Find $\lim_{x \rightarrow -4} (-3x^2 + 8x + 3)$.

a. -81

b. -73

c. -77 ✓

d. 77

$$= -3(-4)^2 - 32 + 3 = -48 - 32 + 3 = -77.$$

- 12 Find $\lim_{x \rightarrow -5} (-3x^2 - 6x + 2)$.

a. -43 ✓

b. -39

c. -47

d. 43

$$= -3(-5)^2 + 30 + 2 = -75 + 30 + 2 = -43.$$

13 Find $\lim_{x \rightarrow -1} (-x^2 - x - 5)$.

a. -1

b. 5

c. -5 ✓

d. -9

$$= -(-1)^2 + 1 - 5 = -1 + 1 - 5 = -5.$$

14 Find $\lim_{x \rightarrow -3} (x^2 - 4x + 5)$.

a. 22

b. 26 ✓

c. -26

d. 30

$$= (-3)^2 + 12 + 5 = 9 + 12 + 5 = 26.$$

15 Find $\lim_{x \rightarrow 5} (-x^2 + 9x - 6)$.

a. -14

b. 18

c. 10

d. 14 ✓

$$= -(5)^2 + 45 - 6 = -25 + 45 - 6 = 14.$$

16 Find $\lim_{x \rightarrow -3} (2x^2 + 5x - 1)$.

a. -2

b. 6

c. 8

d. 2 ✓

$$= 2(-3)^2 - 15 - 1 = 18 - 15 - 1 = 2.$$

17 Find $\lim_{x \rightarrow 0} (3x^2 + 7x + 4)$.

a. 8

b. -4

c. 0

d. 4 ✓

$$= 3(0)^2 + 4 = 0 + 4 = 4.$$

18 Find $\lim_{x \rightarrow 4} (2x^2 + 5x + 9)$.

a. 57

b. -61

c. 65

d. 61 ✓

$$= 2(4)^2 + 20 + 9 = 32 + 20 + 9 = 61.$$

19 Find $\lim_{x \rightarrow -2} (-2x^2 + 6x + 9)$.

a. -7

b. -11 ✓

c. 11

d. -15

$$= -2(-2)^2 - 12 + 9 = -8 - 12 + 9 = -11.$$

20 Find $\lim_{x \rightarrow 5} (3x^2 + x + 8)$.

a. -88

b. 92

c. 88 ✓

d. 84

$$= 3(5)^2 + 5 + 8 = 75 + 5 + 8 = 88.$$

21 Find $\lim_{x \rightarrow -5} (-2x^2 - 8x + 4)$.

a. 6

b. -10

c. -6 ✓

d. -2

$$= -2(-5)^2 + 40 + 4 = -50 + 40 + 4 = -6.$$

22 Find $\lim_{x \rightarrow 1} (x^2 + 4x - 3)$.

a. -2

b. 2 ✓

c. 8

d. 6

$$= (1)^2 + 4 - 3 = 1 + 4 - 3 = 2.$$

23 Find $\lim_{x \rightarrow -2} (-2x^2 + 5x - 9)$.

a. 27

b. -23

c. -27 ✓

d. -31

$$= -2(-2)^2 - 10 - 9 = -8 - 10 - 9 = -27.$$

24 Find $\lim_{x \rightarrow -5} (x^2 - 4x + 9)$.

a. 50

b. 54 ✓

c. -54

d. 58

$$= (-5)^2 + 20 + 9 = 25 + 20 + 9 = 54.$$

Multiple choice — medium

25 Find $\lim_{x \rightarrow -7} \frac{x^2 + 15x + 56}{x + 7}$.

a. 1 ✓

b. -2

c. -1

d. 4

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 15x + 56}{x + 7} = \frac{(x + 7)(x + 8)}{x + 7} = x + 8$ (for $x \neq -7$), so the limit is $(-7) + 8 = 1$.

26 Find $\lim_{x \rightarrow -3} \frac{x^2 + 2x - 3}{x + 3}$.

a. -4 ✓

b. 4

c. -1

d. -7

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 2x - 3}{x + 3} = \frac{(x + 3)(x - 1)}{x + 3} = x - 1$ (for $x \neq -3$), so the limit is $(-3) - 1 = -4$.

27 Find $\lim_{x \rightarrow -5} \frac{x^2 + 6x + 5}{x + 5}$.

a. -1

b. -7

c. -4 ✓

d. 4

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 6x + 5}{x + 5} = \frac{(x + 5)(x + 1)}{x + 5} = x + 1$ (for $x \neq -5$), so the limit is $(-5) + 1 = -4$.

28 Find $\lim_{x \rightarrow -5} \frac{x^2 + 7x + 10}{x + 5}$.

a. 0

b. 3

c. -3 ✓

d. -6

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 7x + 10}{x + 5} = \frac{(x + 5)(x + 2)}{x + 5} = x + 2$ (for $x \neq -5$), so the limit is $(-5) + 2 = -3$.

29 Find $\lim_{x \rightarrow 5} \frac{x^2 - 11x + 30}{x - 5}$.

a. 1

b. -1 ✓

c. 2

d. -4

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 11x + 30}{x - 5} = \frac{(x - 5)(x - 6)}{x - 5} = x - 6$ (for $x \neq 5$), so the limit is $5 - 6 = -1$.

30 Find $\lim_{x \rightarrow -3} \frac{x^2 + 10x + 21}{x + 3}$.

a. 1

b. 7

c. -4

d. 4 ✓

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 10x + 21}{x + 3} = \frac{(x + 3)(x + 7)}{x + 3} = x + 7$ (for $x \neq -3$), so the limit is $(-3) + 7 = 4$.

31 Find $\lim_{x \rightarrow 2} \frac{x^2 - 9x + 14}{x - 2}$.

a. -5 ✓

b. -2

c. -8

d. 5

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 9x + 14}{x - 2} = \frac{(x - 2)(x - 7)}{x - 2} = x - 7$ (for $x \neq 2$), so the limit is $2 - 7 = -5$.

32 Find $\lim_{x \rightarrow 6} \frac{x^2 - 14x + 48}{x - 6}$.

a. -5

b. 2

c. -2 ✓

d. 1

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 14x + 48}{x - 6} = \frac{(x - 6)(x - 8)}{x - 6} = x - 8$ (for $x \neq 6$), so the limit is $6 - 8 = -2$.

33 Find $\lim_{x \rightarrow -5} \frac{x^2 - 25}{x + 5}$.

- a. 10
- b. -13
- c. -10 ✓
- d. -7

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 25}{x + 5} = \frac{(x + 5)(x - 5)}{x + 5} = x - 5$ (for $x \neq -5$), so the limit is $(-5) - 5 = -10$.

34 Find $\lim_{x \rightarrow 7} \frac{x^2 - 4x - 21}{x - 7}$.

- a. 13
- b. -10
- c. 7
- d. 10 ✓

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 4x - 21}{x - 7} = \frac{(x - 7)(x + 3)}{x - 7} = x + 3$ (for $x \neq 7$), so the limit is $7 + 3 = 10$.

35 Find $\lim_{x \rightarrow -2} \frac{x^2 - 2x - 8}{x + 2}$.

- a. -6 ✓
- b. -3
- c. 6
- d. -9

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 - 2x - 8}{x + 2} = \frac{(x + 2)(x - 4)}{x + 2} = x - 4$ (for $x \neq -2$), so the limit is $(-2) - 4 = -6$.

36 Find $\lim_{x \rightarrow -3} \frac{x^2 + x - 6}{x + 3}$.

- a. -2
- b. -8
- c. -5 ✓
- d. 5

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + x - 6}{x + 3} = \frac{(x + 3)(x - 2)}{x + 3} = x - 2$ (for $x \neq -3$), so the limit is $(-3) - 2 = -5$.

37 Find $\lim_{x \rightarrow -3} \frac{x^2 + 9x + 18}{x + 3}$.

- a. 6
- b. 3 ✓
- c. -3
- d. 0

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 9x + 18}{x + 3} = \frac{(x + 3)(x + 6)}{x + 3} = x + 6$ (for $x \neq -3$), so the limit is $(-3) + 6 = 3$.

38 Find $\lim_{x \rightarrow 1} \frac{x^2 + 6x - 7}{x - 1}$.

- a. 5
- b. -8
- c. 8 ✓
- d. 11

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 6x - 7}{x - 1} = \frac{(x - 1)(x + 7)}{x - 1} = x + 7$ (for $x \neq 1$), so the limit is $1 + 7 = 8$.

39 Find $\lim_{x \rightarrow 0} \frac{x^2 + 8x}{x}$.

a. 11

b. 8 ✓

c. 5

d. -8

Substitution gives $\frac{0}{0}$. Factor: $\frac{x^2 + 8x}{x} = \frac{(x)(x + 8)}{x} = x + 8$ (for $x \neq 0$), so the limit is $0 + 8 = 8$.

Multiple choice — hard

40 Given $f(x) = -x^2 + 5x - 4$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(2)$.

a. 1 ✓

b. -1

c. 3

d. 4

$f'(x) = -2x + 5$, so $f'(2) = -2(2) + 5 = -4 + 5 = 1$.

41 Given $f(x) = 4x^2 + 5x + 9$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(-6)$.

a. -45

b. 43

c. -41

d. -43 ✓

$f'(x) = 8x + 5$, so $f'(-6) = 8(-6) + 5 = -48 + 5 = -43$.

42 Given $f(x) = -3x^2 + 9x - 2$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(3)$.

- a. -7
- b. -11
- c. 9
- d. -9 ✓

$$f'(x) = -6x + 9, \text{ so } f'(3) = -6(3) + 9 = -18 + 9 = -9.$$

43 Given $f(x) = 4x^2 + 3x + 7$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(4)$.

- a. 35 ✓
- b. -35
- c. 33
- d. 37

$$f'(x) = 8x + 3, \text{ so } f'(4) = 8(4) + 3 = 32 + 3 = 35.$$

44 Given $f(x) = -x^2 - 3x - 1$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(0)$.

- a. -1
- b. -5
- c. -3 ✓
- d. 3

$$f'(x) = -2x - 3, \text{ so } f'(0) = -2(0) - 3 = 0 - 3 = -3.$$

- 45 Find $\lim_{x \rightarrow \infty} \frac{6x + 3}{x - 9}$.
- a. $\frac{13}{2}$
 - b. -6
 - c. $\frac{11}{2}$
 - d. 6 ✓

Divide numerator and denominator by the highest power of x (here x): as $x \rightarrow \infty$, the constant terms vanish, leaving $\frac{6}{1} = 6$.

- 46 Find $\lim_{x \rightarrow \infty} \frac{3x + 1}{-5x - 6}$.
- a. $-\frac{1}{10}$
 - b. $\frac{3}{5}$
 - c. $-\frac{11}{10}$
 - d. $-\frac{3}{5}$ ✓

Divide numerator and denominator by the highest power of x (here x): as $x \rightarrow \infty$, the constant terms vanish, leaving $\frac{3}{-5} = -\frac{3}{5}$.

- 47 Find $\lim_{x \rightarrow \infty} \frac{x + 5}{3x - 8}$.
- a. $\frac{5}{6}$
 - b. $\frac{1}{3}$ ✓
 - c. $-\frac{1}{3}$
 - d. $-\frac{1}{6}$

Divide numerator and denominator by the highest power of x (here x): as $x \rightarrow \infty$, the constant terms vanish, leaving $\frac{1}{3}$.

- 48 Find $\lim_{x \rightarrow \infty} \frac{-2x}{-5x + 2}$.
- a. $-\frac{2}{5}$
 - b. $\frac{2}{5}$ ✓
 - c. $\frac{9}{10}$
 - d. $-\frac{1}{10}$

Divide numerator and denominator by the highest power of x (here x): as $x \rightarrow \infty$, the constant terms vanish, leaving $\frac{-2}{-5} = \frac{2}{5}$.

- 49 Find $\lim_{x \rightarrow \infty} \frac{-4x - 2}{-4x + 4}$.
- a. $\frac{3}{2}$
 - b. 1 ✓
 - c. $\frac{1}{2}$
 - d. -1

Divide numerator and denominator by the highest power of x (here x): as $x \rightarrow \infty$, the constant terms vanish, leaving $\frac{-4}{-4} = 1$.

- 50 Find the average rate of change of $f(x) = x^2$ on the interval from $x = -1$ to $x = 0$.
- a. -3
 - b. 3
 - c. 1
 - d. -1 ✓

$$\frac{f(0) - f(-1)}{0 - (-1)} = \frac{0^2 - (-1)^2}{1} = \frac{-1}{1} = -1.$$

51 Find the average rate of change of $f(x) = x^2$ on the interval from $x = 3$ to $x = 1$.

a. 2

b. 6

c. -4

d. 4 ✓

$$\frac{f(1) - f(3)}{1 - 3} = \frac{1^2 - 3^2}{-2} = \frac{-8}{-2} = 4.$$

52 Find the average rate of change of $f(x) = x^2$ on the interval from $x = -1$ to $x = 1$.

a. 4

b. -2

c. 2

d. 0 ✓

$$\frac{f(1) - f(-1)}{1 - (-1)} = \frac{1^2 - (-1)^2}{2} = \frac{0}{2} = 0.$$

53 Find the average rate of change of $f(x) = x^2$ on the interval from $x = 3$ to $x = 4$.

a. 5

b. -7

c. 9

d. 7 ✓

$$\frac{f(4) - f(3)}{4 - 3} = \frac{4^2 - 3^2}{1} = \frac{7}{1} = 7.$$

54 Find the average rate of change of $f(x) = x^2$ on the interval from $x = 2$ to $x = -1$.

a. 3

b. 5

c. 1 ✓

d. -1

$$\frac{f(-1) - f(2)}{-1 - 2} = \frac{(-1)^2 - 2^2}{-3} = \frac{-3}{-3} = 1.$$